NTNU

Faculty of Information Technology and Electrical Engineering (IE)
Department of Electric Power Engineering

Learning Objectives and Study Guide TET4185, Power Markets, Resources and Environment

Spring 2023

In general, I will often expand somewhat on the material presented in the book. This is related to updates, additional material or material I want to present in a different way. In addition to the book, all lecture slides are also part of the curriculum.

In the following, I will very shortly present the major issues in each lecture. What is written here is certainly exam relevant (which does not mean that you can forget everything else for the exam).

Chapter 1, Introduction

This chapter introduces the course and gives some background information. Important issues are the traditional organisation and the major motivation for it. In the lecture, I will talk about the reasons for restructuring. Important is also the special features of electricity that make this market special. I will also go into the main features of a good market design regarding short-term operation and talk about investment incentives. This is a very important issue in electricity market design – without new investments, reliability will ultimately become unacceptable. I will also talk about the major design features of the wholesale market and shortly mention two very different but well-functioning market designs. Finally, I will discuss criticism of deregulation.

Learning objectives:

- The main reason for the historical organisation of the power sector
- The reason why the power sectors in many countries were restructured from 1990 and onwards
- The special features of electricity
- What characterises a good market design
- What is meant by "investment incentives" and why this is important
- Criticism of deregulation/restructuring

Chapter 2, Basic microeconomics

This chapter is a short update on microeconomics and contains material that is part of the basis for this course. It is an advantage if you already know this, but if not, the material in the book and what is presented in the lecture is sufficient for the remainder of the course.
We start with a short repetition of the marginal utility of demand and the marginal cost of production and the fact that social welfare is maximised in the Marshallian demand cross, where the marginal utility of demand is equal to the marginal cost of production. This is maybe the fundamental insight of microeconomics, and it is also extremely important in this course. If you feel uncertain about the definition of consumer and producer surplus and social welfare, make sure to look it up!
We will spend a little time on the concept of demand elasticity. We will only define demand elasticity and some specific forms of it (direct, cross, income). We will not go into different forms of demand

curves, and in examples we will (as is done in many scientific papers) normally assume linear demand curves (although these are not very realistic).
We will then look into various forms of competition, and do some basic derivations of the major rules. With respect to oligopolistic competition, we will focus on Cournot competition in this course, but you should be aware that there are many theoretical models for this.
At the end of the lecture, I will give a small power system example of the three major forms of competition. There will also come an exercise about this. This example is not in the book.

Learning objectives

- Supply-demand cross, social welfare, consumer and producer surplus
- Definition and use of demand elasticity
- The relation between total cost, average cost and marginal cost
- Market behaviour and price for the three major kinds of competition:
  - perfect
  - monopolistic
  - oligopolistic (Cournot)
- Be able to calculate quantities and prices in simple examples of non-perfect competition
- Have a principal understanding of the effect of taxes on the market equilibrium

Chapter 3, Electricity Consumption

This chapter is out of scope for this course. Instead, we selected chapter 3 and chapter 4 from "Fundamentals of Power System Economics" by Daniel S. Kirschen and Goran Strbac, 2004.
This lecture is rather broad, including background information that you should know about power system economics and different types of trading. In general, bilateral trading involves only two parties: a buyer and a seller. Participants thus enter into contracts without involvement, interference or facilitation from a third party. Since electrical energy is pooled as it flows from the generators to the loads, it was felt that trading might as well be done in a centralised manner and involve all producers and consumers. Competitive electricity pools were thus created. Rather than relying on repeated interactions between suppliers and consumers to reach the market equilibrium, a pool provides a mechanism for determining this equilibrium in a systematic way. Furthermore, different market design, including day-ahead, intra-day, and the balancing market, will be explained. Based on the time horizon of the bids and the time span between the gate closure time and actual energy delivery time, Power exchanges perform competitive trading in three types of auctions:

- Day-Ahead (DA).
- Intra-Day (ID).
- Balancing Market

Following up, we discuss the decisions that generators, consumers and others take to optimise the benefits that they derive from these markets. We will look at the consumer's perspective, the retailer of electrical energy. We will first discuss why consumers have a much more passive role than producers do in electricity markets and how retailers serve as their intermediaries in the market for electrical energy. We will then adopt the perspective of a generating company and consider the case in which this company faces a perfectly competitive market.

Learning objectives

- Different types of electricity trading
- The time-line of the electricity market in the Nordic power market
- Participating in electricity markets: the consumer's perspective
- Options for the consumers
- Participating in electricity markets: the producer's perspective
- Marginal, infra-marginal, extra-marginal producers
- Bidding under perfect competition
- Optimal scheduling, start-up cost, temporal constraints and environmental constraints

Chapter 4, Pricing

Naturally, pricing is a major issue in a power market, and it is not possible to go into all aspects within the timeframe of this course. Important to learn from this chapter and the lecture is the criteria used to determine the tariffs, the effect of unbundling (more about that in the next chapter), common types of contracts and tariffs and their characteristics and marginal cost pricing.
Unbundling means splitting up the power sector in generation, transmission and distribution and retail of supply. This is a process done in some countries during the 1990s, but still an ongoing process in many other places. This also leads to an explicit corresponding split-up of the bills of the consumers.
It is important to know the theoretical basis for marginal cost pricing and the practical forms of implementation. Furthermore, the effects of pricing according to short and long-run marginal cost. There will be an exercise about this. Finally, we will discuss Ramsey pricing, why it is beneficial for society and how it is used.

Learning objectives:

- Know the major criteria for the determination of tariffs and power contracts
- Know the common types of contracts and their characteristics, advantages and disadvantages
- Know and understand the major contract types in Norway
- Be able to calculate the consequences (producer and consumer surplus, social welfare) of setting prices according to SRMC or LRMC
- Understand Ramsey pricing and be able to do simple calculations with this method

Chapter 5, Restructuring

Chapter 5 is very broad and discusses many sides of power system restructuring, and it may be a bit hard to know "what should I know about this". In general, you should know about the major issues that are important in a "restructured" power system. In fact, these are the elements that should be included in a good market design.

You should understand and be able to explain the roles of the different market actors. The difference between a power pool like PJM and a power exchange like Nord Pool and the meaning of "portfolio bidding" is also important. Furthermore, the difference between energy-only markets and capacity markets and why some markets have the additional capacity element. A significant part of the chapter is about the Nordic and Norwegian market. This is important in its own right, because Nord Pool (as it is often called) is often cited as an example of a successful market implementation, and because most of you are Norwegian or will work in Norway, and as such should know the basics of this market. You should also understand the roles of the different markets in the Nord Pool system: financial, spot (day ahead), intraday, balancing.

Learning objectives:

- Understand the reasons behind restructuring
- Know the major roles in the power market
- Know the major elements of market design
- Know the difference between energy only and capacity markets, and the reason for capacity mechanisms
- Know the major functioning of power pools and exchanges
- Understand the different markets and their roles (financial, spot, intraday, balancing)
- Know the major characteristics of the Nordic market

Chapter 6, Generation

This chapter is rather broad, and it may not be obvious what you should know well and what is "background" information. Chapters 6.2 and 6.3 are repletion from TET4135, Energy system planning and operation, and will not spend much time on this. Of course, you are assumed to understand and be able to use these approaches analytically. If you do not, look back a little on TET4135. In the lecture, I will briefly discuss the different thermal and hydro units' properties and what impact they have on markets.

The bathtub model is another way to explain water values and to illustrate the impact of constraints. You should understand this approach and be able to do calculations on simple models. In the lecture, I will let you work a little bit on such a model.

Then there is a description of hydro scheduling, based on the scenario tree for possible future market price. The remainder of this chapter is not included in this course.

Finally, you should be able to use the approach in Section 6.8 to find the optimal generation mix in a simple system, which is a basis for the lecture on the fundamental of the capacity market.

Learning objectives:

- Understand and be able to apply simple dispatch of units with given resource costs (repetition)
- Understand the bathtub model and be able to do simple calculations based on the model
- Be able to use a simple price model for hydro scheduling based on the scenario tree for possible future market price
- Understand and be able to apply the simple screening model of Section 6.7 for the optimal generation mix

Chapter 7, Grid access

This is the central chapter of the course. It contains the following main issues:

7.2 – Congestion management using area pricing and buy-back (or the more usual term counter trading)
7.3 and Appendix – Congestion management through nodal pricing
7.4 Transmission and distribution tariffs

In the present course, 7.5 is not included.

Generally speaking, "congestion management" is about the handling of constraints or bottlenecks in the grid. In a general market context, all consumers will, in theory, have the same access to the traded goods (although there are many exceptions on this in most markets). In the power market, access to power is restricted by the grid. If I live in an area with high prices and want to buy power in an area with low prices, this is not possible because of the grid constraints (if there were no grid constraint, there would be no price difference). Congestion management is basically about the handling of physical grid constraints in the market context.

With respect to area pricing, you should understand how to calculate producer and consumer surplus in an isolated area. It would help if you also understood how these change when the area is connected with another area with a different price and how this increases the total surplus, made up of consumer surplus, producer surplus and congestion rent. You should also know about the welfare loss caused by the fact that there is insufficient exchange capacity to make prices in the areas equal. You should be able to do calculations of this on simple examples. The next section explains the buy-back procedure, how it works and how it influences costs and revenues for the involved parties. You should understand this, the difference with area pricing and advantages and disadvantages with each method.

The use of Optimal Power Flow (OPF) is an alternative way to handle transmission constraints. While area pricing is used in the Nordic countries and in Europe between (but mostly not within) countries, the use of Nodal Pricing based on OPF is used in several places in the USA. Nodal prices based on OPF are a more physically correct way to represent the grid constraints, but at the cost of different prices in each node in the network. Sections 7.3.1 and 7.3.2 explain the overall principles, while 7.3.3 explains the handling of losses. The Appendix describes the mathematics in much more detail for the small 3-node example that is introduced in Section 7.3.8. The recommended approach is:

1. Read the memo on Lagrange multipliers as a freshening up of (hopefully) existing knowledge
2. Read 7.3.1 and 7.3.2, then quickly look at the example in 7.3.8
3. Go through the Appendix until you come to the part about losses
4. Read 7.3.3 about losses
5. Continue and finish the Appendix
6. Read the memo "DC Power Flow in Power Markets"

If you understand all this well and can do the exercises, you have really taken the important issues! After this, it should not be a problem to understand Sections 7.3.4-7.3.7 and the remainder of 7.3.8. Instead of the three-node DC equivalent in Section 7.3.9 we use the memo "DC Power Flow in Power Markets". Note that only the part from "DC Power Flow" is curriculum, but the first part is a basis for understanding the approach here. Section 7.3.10 is not part of this year's curriculum. Section 7.3.11 illustrates the OPF approach with an 11-node network. There is also an illustration using the Nordic grid in the lecture notes. It is recommended to go through the lecture notes; they give some additional examples, explicit calculations and corrections of errors in the book.

Section 7.4 is about principles for grid tariffs and, more specifically, the Norwegian and Swedish systems. Finally, there is little material about market coupling, the current approach to coupling the markets in European countries.

Learning objectives:
Section 7.2:

- Understand calculation of producer and consumer surplus in a single area
- Understand how these are influenced by the interconnection of areas and how this affects the total surplus through the producer and consumer surpluses and the congestion rent
- Understand the loss of social welfare caused by insufficient interconnection capacity
- Understand the principles of counter-trade of buy-back
- Be able to solve simple cases of area pricing and counter trades in practice
- Understand the advantages and disadvantages of each method

Section 7.3, Appendix and memo "DC Power Flow in Power Markets"

- Understand the meaning of the Lagrangian function representing the production cost minimisation and the constraints that result from the load flow equations, g(x,P) = 0, as well as the grid constraints, h(x) = 0, and the constraints on maximum generation
- Be able to solve this problem for small examples both for a DC grid and a DC load flow in an AC grid
- Understand how the power prices are calculated in the OPF context
- Understand the calculation of the loss factors and how they are used to calculate the grid tariffs
- Understand how constraints affect prices in an OPF – nodal prices context

Section 7.4

- Know the main principles for grid tariffing in Norway and Sweden
- Understand the principle of market coupling and how it is applied in CWE – Nordic

Chapter 7, Flow-Based Market Coupling

Flow-based market coupling builds on the DC Optimal Power Flow theory and is the presently preferred market-clearing method in the European Union. It can be viewed as a compromise between nodal pricing, with a detailed (though still algorithmically limited) representation of the grid, and area

pricing (also called market splitting), which represents only the major limitations in the grid without considering load flow constraints.
Area pricing, or the ATC based method as it is normally called in the European context, sets fixed limits on the exchange between the price areas (Bidding Zones), not considering the interdependence between the limits. In the Flow-Based Approach, on the other hand, the simplified inclusion of the load flow equations leads to an implicit dependence between the flows.
The Power Transfer Distribution Factors (PTDFs) constitute one of the key concepts of the method. They describe how a change in flow between two areas translates into changes flows on the Critical Network Elements (CNEs), which are the connections between the areas as well as a number of other major lines and transformers.
Another key concept is the Flow-Based Domain, which is a shape in the (n-1)-dimensional room describing the constraints for a system with n areas. Any market-clearing solution must lie within the Flow-Based Domain, and the objective of the algorithm is to find the solution that maximises the economic surplus within this domain. The Flow-Based Domain is larger than the ATC based domain because it can handle simultaneous constraints in a better way.
Aggregation is central to the Flow-Based method. The PTDFs are first calculated for the full grid but need subsequently be aggregated per area. This is done by using the Generation Shift Keys that describe how a change in an areas net position (net export minus import) translates into changes in the injections in the individual nodes.

Learning Objectives:

- Understand and be able to explain the basics of the Flow-Based method and its motivation
- Understand the calculation of the PTDFs and be able to use the PTDFs to calculate inter-area flows
- Understand the concept of the Flow-Based Domain and the difference with the ATC domain, and be able to calculate the Flow-Based Domain for simple cases
- Understand the Generation Shift Keys and their application

Chapter 8, Ancillary services
Ancillary services (AS) are "Services necessary for the operation of an electric power system provided by the system operator and/or by power system users". AS are necessary to maintain the security of supply (SoS), which to a large extent can be viewed as a collective or public good, that is not supplied by a market (e.g. compare street lighting or a country's defence). Therefore, it is the role of the TSO as the responsible party for SoS to provide ancillary services.

A classification of AS is given in Table 8.1 in the book. However, many developments are presently going on in this area in Europe, with the main objective of harmonising ancillary services and enabling better market integration to lower costs. These processes are not reflected in the book but are described in the lecture notes. Especially important is the reference model defining Frequency Controlled Reserves (FCR), Frequency Restoration Reserves (FRR) and Replacement Reserves (RR). For FCR both the book and the lecture notes define the *unit droop* and how it is used.

Subsequently, the lecture notes give the relation of the reference model with the current Nordic and RGCE definitions and models that largely (but not fully) correspond to Tables 8.2 and 8.3 in the book. It is not necessary to remember exact values for all areas, but you should know the meaning of the main operational reserve parameters.

Some of the services are traded in markets with the TSO as a single buyer. This is also described in the lecture notes and for the Nordic system in Sections 8.8 and 8.9.

Learning objectives:

- Know the classification of ancillary services
- Know and understand the reference model (FCR, FRR, RR)
- Understand and be able to use unit droop in simple calculations
- Understand the main operational reserve parameters
- Know the mapping of the reference model on present the Nordic system
- Know the major markets in Norway and the Nordic system (FNR/FDR, RK, ROM)
- Understand the simple model to calculate the opportunity cost of reserves

Chapter 10, Environmental Policy

This chapter discusses some aspects of environmental policy and specific instruments to reduce emissions. Apart from a general understanding, you should know some of the major instruments that are used today, mentioned in Slide 12 of the lecture slides. Moreover, you should understand and be able to outline the major characteristics of the EU-ETS.

The final part of the chapter describes the Green Certificate or TGS system in a welfare economics context with a focus on prices and social welfare. You should understand the arguments in the various figures (10.11 – 10.20) and do simple calculations, as given in Assignment 8.

Learning objectives

- Understand the meaning of external costs
- Know the main policies to reduce emissions
- Know the main characteristics of ETS
- Understand and be able to explain the Green Certificate system
- Be able to do price calculations based on simple models like those given in Figures 10.10 – 10.20 in the book